

MOLECULAR BASIS OF INHERITANCE

Comprehensive Analytical Study Material & Master Exam Reference Guide

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1. Structural Architecture and Chemistry of DNA

Deoxyribonucleic Acid (DNA) is a long linear polymer consisting of repeated deoxyribonucleotide sub-units. The specific length and base sequence represent the genetic blueprint of an organism.

Standard Genome Lengths

- **phi-X-174 Bacteriophage:** 5,386 nucleotides (ssDNA)
- **Bacteriophage Lambda (λ):** 48,502 base pairs (bp)
- **Escherichia coli:** 4.6×10^6 bp (~1.36 mm length)
- **Human Genome (Haploid, n):** 3.3×10^9 bp
- **Human Genome (Diploid, 2n):** 6.6×10^9 bp (~2.2 meters)

Chemical Components

- **Purines (double-ringed):** Adenine (A), Guanine (G)
- **Pyrimidines (single-ringed):** Cytosine (C), Thymine (T) in DNA, Uracil (U) in RNA
- **Pentose Sugar:** 2'-deoxyribose (lacks oxygen at 2' C)
- **Phosphate Group:** Imparts negative charge and acidity

Key Covalent and Non-Covalent Bonds

- **N-Glycosidic Bond:** Formed between Nitrogen-1 of pyrimidine or Nitrogen-9 of purine and Carbon-1' of the pentose sugar to form a *nucleoside*.
- **Phosphoester Bond:** Formed between the 5'-OH group of the pentose sugar and a phosphate group to form a *nucleotide*.
- **3' - 5' Phosphodiester Bond:** Covalently joins the 3'-OH group of one nucleotide's pentose sugar to the 5'-phosphate group of the adjacent nucleotide.
- **Hydrogen Bonds:** Maintain complementary base-pairing (A = T with 2 H-bonds; G $\equiv$ C with 3 H-bonds).

The Watson-Crick B-DNA Double Helix Model (1953)

Proposed based on X-ray diffraction data by **Maurice Wilkins & Rosalind Franklin**, alongside the base equivalence rules discovered by **Erwin Chargaff**.

Chargaff's Rules of Base Equivalence (For Double-Stranded DNA)

- Total Purines = Total Pyrimidines: $[A] + [G] = [T] + [C]$
- Stoichiometric Ratios: $[A]/[T] = 1$ and $[G]/[C] = 1$
- Base Ratio $([A] + [T]) / ([G] + [C])$ is species-specific and remains constant for an organism.

Parameter	Quantitative / Structural Dimension
Helical Geometry	Two anti-parallel strands (5' → 3' and 3' → 5') coiled in a right-handed helix.
Helix Diameter	2.0 nm (20 Å) — constant due to uniform Purine-Pyrimidine pairing.
Pitch of Helix	3.4 nm (34 Å) per full 360° helical turn.
Base Pairs per Turn	10 bp per turn; Axial rise per base pair = 0.34 nm (3.4 Å) .
Stability Factors	Hydrogen bonding between complementary bases + hydrophobic base-stacking interactions.

2. Packaging of DNA Helix

Packaging Problem & Quantitative Derivation

Human diploid cell: 6.6×10^9 bp $\times$ 0.34×10^{-9} m/bp $\approx$ **2.2 meters**. This entire length must be tightly condensed inside a microscopic nucleus ($\sim 10^{-6}$ m diameter).

- **Prokaryotes (e.g., E. coli):** Lack membrane-bound nuclei. The 1.36 mm circular DNA is organized into large supercoiled loops within the **nucleoid**, held by basic polyamines and RNA/protein complexes.
- **Eukaryotes (Nucleosome Organization):**
 - **Histone Octamer:** Core composed of two molecules each of four basic proteins (H2A, H2B, H3, H4), rich in positively charged amino acids **Lysine** and **Arginine**.
 - **Nucleosome Core Particle:** ~ 200 bp of negatively charged DNA wound ~ 1.65 turns around the histone octamer.
 - **H1 Linker Histone:** Binds entry and exit points, sealing the wrapped core.
 - **Chromatin Hierarchy:** Nucleosomes ("beads-on-a-string", 11 nm) → Solenoid fiber (30 nm) → Looped domains → Metaphase chromosome (anchored by Non-Histone Chromosomal / NHC proteins).

Feature	Euchromatin	Heterochromatin
Chromatin Packing	Loosely packed, stains light	Densely condensed, stains dark
Transcriptional State	Transcriptionally Active	Transcriptionally Inactive (Silent)
Replication Timing	Replicates early during S phase	Replicates late during S phase

3. Experimental Proof of DNA as the Genetic Material

1. Griffith's Transformation Experiment (1928)

Injected *Streptococcus pneumoniae* (virulent Smooth 'S' with capsule vs non-virulent Rough 'R' without capsule) into mice:

- Live S strain → Mouse dies | Live R strain → Mouse lives
- Heat-killed S strain → Mouse lives
- Live R + Heat-killed S → Mouse dies (Live virulent S bacteria recovered from dead mouse).
- **Conclusion:** A "transforming principle" transferred genetic instructions from heat-killed S to R strain.

2. Biochemical Characterization (Avery, MacLeod, & McCarty, 1933-44)

- Treated heat-killed S extracts with Proteases (no effect on transformation).
- Treated with RNase (no effect on transformation).
- Treated with **DNase** (completely eliminated transforming capability).
- **Conclusion:** DNA is the chemical substance responsible for transformation.

3. Hershey-Chase Blender Experiment (1952) — Unequivocal Proof

Bacteriophage T2 labeled with radioisotopes and infected into *E. coli*:

- ^{35}S labeled viral protein coat → After agitation & centrifugation, radioactivity remained in **supernatant**.
- ^{32}P labeled viral DNA → After centrifugation, radioactivity was detected exclusively in the **bacterial pellet**.
- **Conclusion:** DNA, not protein, enters host cells and directs viral progeny synthesis.

Analytical Comparison: DNA vs. RNA

Parameter	DNA	RNA
Chemical Nature	2'-deoxyribose (lacks 2'-OH group) → highly unreactive, stable.	Ribose has a reactive 2'-OH group → makes RNA labile, catalytic, easily degraded.
Thymine vs. Uracil	Contains Thymine (5-methyluracil) conferring higher stability.	Contains Uracil → susceptible to oxidative deamination.
Biological Function	Optimal for long-term storage of genetic instructions.	Optimal for transmission, catalysis, and expression (tRNA, rRNA, mRNA).

4. DNA Replication

Replication is **semi-conservative** (Watson-Crick, 1953): each newly synthesized double-stranded DNA contains one parental template strand and one newly formed complementary strand.

Meselson and Stahl Experiment (1958) on *E. coli*

- Grown in $^{15}\text{NH}_4\text{Cl}$ medium → Pure heavy DNA (^{15}N - ^{15}N) formed.
- Transferred to ^{14}N medium and analyzed across generations by **CsCl density gradient centrifugation**:
 - **Generation I (20 min):** 100% Hybrid density DNA (^{15}N - ^{14}N).
 - **Generation II (40 min):** 50% Light (^{14}N - ^{14}N) and 50% Hybrid (^{15}N - ^{14}N).
- *Similar validation on *Vicia faba* chromosomes was achieved by Taylor et al. (1958) using ^3H -thymidine.*

Replication Machinery & Fork Mechanism

- **Origin of Replication (ori):** Designated initiation locus.
- **DNA Helicase & Topoisomerase:** Unwinds double helix and relieves positive torsional strain.
- **Deoxyribonucleoside Triphosphates (dNTPs):** Serve dual functions: substrates + high-energy phosphate donors.
- **DNA-dependent DNA Polymerase:** Synthesizes exclusively in the **5' → 3'** direction with proofreading capability.
- **Leading vs. Lagging Strand:**
 - *Leading strand:* Continuous synthesis along the 3' → 5' template toward the replication fork.
 - *Lagging strand:* Discontinuous synthesis along the 5' → 3' template producing **Okazaki fragments**, sealed covalently by **DNA Ligase**.

5. Transcription Mechanism & Operons

3' — [Promoter] — [Structural Gene] — [Terminator] — 5' (Template Strand)
5' — [Promoter] — [Structural Gene] — [Terminator] — 3' (Coding Strand)

Transcription is asymmetric: only one DNA strand (the **3' → 5' template strand**) is transcribed into RNA. Promoter and terminator coordinates are defined relative to the **coding strand** (5' upstream, 3' downstream).

Prokaryotic vs Eukaryotic Transcription

Feature	Prokaryotes	Eukaryotes
Enzymatic Diversity	Single RNA Polymerase + σ (sigma) initiation factor & ρ (rho) termination factor.	Three distinct RNA Polymerases: • Pol I: 28S, 18S, 5.8S rRNA • Pol II: hnRNA (mRNA precursor) • Pol III: tRNA, 5S rRNA, snRNA
Gene Organization	Polycistronic operons (multiple genes under one promoter).	Monocistronic split genes with non-coding <i>introns</i> and coding <i>exons</i> .
Post-Transcriptional Processing	None; translation occurs cotranscriptionally.	Splicing (intron removal), 5' Capping (m^7Gppp), and 3' Polyadenylation (200-300 poly-A tail).

6. Genetic Code, tRNA & Protein Translation

Salient Features of the Genetic Code

- **Triplet Codon:** 64 total codons (61 sense codons + 3 stop codons: UAA, UAG, UGA).
- **Unambiguous & Specific:** One codon codes for exactly one amino acid.
- **Degenerate:** Most amino acids are specified by multiple codons.
- **Universal & Commaless:** Read continuously 5' → 3' without spacing across nearly all taxa.
- **Dual Function of AUG:** Serves as the universal start codon and specifies Methionine (Met).

The tRNA Adapter Molecule

- **Anticodon Loop:** Binds complementary mRNA codons antiparallely.
- **Amino Acid Acceptor Stem:** CCA-3' terminus covalently binds activated amino acid via Aminoacyl-tRNA Synthetase.
- **DHU & T ψ C Loops:** Mediate enzyme recognition and ribosomal binding.
- *Note: No tRNAs exist for stop codons.*

Steps in Translation

1. **Activation & Charging of tRNA:** Amino acid + ATP → Aminoacyl-AMP + PPi → Aminoacyl-tRNA.
2. **Initiation:** Small ribosomal subunit (30S/40S) binds mRNA at start codon (AUG); initiator Met-tRNA pairs at P-site; large subunit joins.
3. **Elongation & Catalysis:** Next aminoacyl-tRNA enters A-site; **23S rRNA in bacteria (28S in eukaryotes)** acts as a catalytic **Ribozyme** to forge peptide bonds. Ribosome moves 5' → 3'.
4. **Termination:** Release factor binds stop codons at A-site, releasing polypeptide chain.

7. Regulation of Gene Expression: The Lac Operon

Elucidated by François Jacob and Jacques Monod (1961), the **Lac Operon** is an inducible prokaryotic system for lactose catabolism.

[p] – [i] – [p] – [o] – [z] – [y] – [a]

Gene	Classification	Analytical Function
i gene	Regulator (Constitutive)	Synthesizes the Lac Repressor protein.
p (Promoter)	Regulatory Site	Binding site for RNA Polymerase.
o (Operator)	Regulatory Switch	Binds Repressor protein to block RNA Polymerase progression.
z gene	Structural Gene	Encodes β-galactosidase (hydrolyzes lactose into glucose + galactose).
y gene	Structural Gene	Encodes β-galactoside permease (increases cell permeability to lactose).
a gene	Structural Gene	Encodes β-galactoside transacetylase .

Negative Regulation (Absence of Inducer)

Active repressor protein binds operator (o), preventing RNA polymerase transcription. Structural gene transcription is repressed.

Induction (Presence of Lactose / Inducer)

Lactose / Allolactose binds repressor, causing conformational inactivation. RNA Polymerase transcribes polycistronic lac mRNA.

8. Human Genome Project (HGP) & DNA Fingerprinting

HGP Core Facts & Figures

- Duration: 1990–2003 (Coordinated by NIH, US DOE, Wellcome Trust).
- Human genome contains **3.1647×10^9 bp** (~30,000 genes).
- Average gene = 3,000 bp; Largest gene = **Dystrophin** (2.4 million bp).
- < 2% of genome codes for proteins.
- Chromosome 1 has the most genes (2,968); Y chromosome has the fewest (231).
- 1.4 million Single Nucleotide Polymorphisms (SNPs) identified.

DNA Fingerprinting (Alec Jeffreys)

- Based on **VNTRs (Variable Number of Tandem Repeats)** / Mini-satellites (0.1–20 kb) exhibiting high allelic polymorphism.
- **Steps:**
 1. DNA isolation & Restriction endonuclease digestion.
 2. Agarose gel electrophoresis.
 3. Southern blotting to nylon membrane.
 4. Hybridization using radiolabeled VNTR probe.
 5. Autoradiography detection of unique band patterns.